OSSEM Test Plan

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Team Members: Hussein Ashkanani, Ethan Budde, David Mazhnikov, Roman Sulza, Clarker Ren

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Top Down Test Plan v1.0

Date of Test:	
Testers:	

Purpose

To test the full espresso emissions system and verify that the espresso machine, emissions chamber, airflow system, sensors, scale, data logging, and wireless communications function correctly before, during, and after a brew cycle.

Equipment Needed

- System Equipment
  - Gaggia Espresso Machine (Gaggiuino Installed)
  - Emissions Control Unit (ECU) / Emissions chamber
  - 3 Fans
    - 2 fans for intake and exhaust ventilation
    - 1 fan for internal air circulation
  - ESP32 Controllers
  - Scale
  - SGP30, SCD30 and BME680 Sensors
  - Power supply to espresso machines (120V) and ECU (3.3V & 12V)
  - Water and coffee beans
  - Coffee Grinder
  - Portafilter and Cup
  - Coffee Tamper
  - Espresso Coffee Stirrer (WDT)

- Test Equipment
  - Laptop or computer for live data display
  - USB cable to power ESP32 for setup/programming
  - Network connection / Hotspot / Wi-Fi access

- Other Equipment
  - Paper Towels / Cleanup materials / Sink
  - Notebook or spreadsheet for logging
  - Safety gloves if needed for hot surfaces (such as portafilter)

Pre-Test Setup

1. Place the espresso machine and ECU in the testing environment.
2. Verify all power connections are secure.
3. Verify the ESP32 boards are powered and programmed.
4. Connect the scale, sensors, fans, and communication modules.
5. Connect the computer to the monitoring interface.
6. Fill the espresso machine water reservoir and prepare the machine for brewing.
7. Weigh 18 grams of coffee beans and grind them in the coffee grinder (using the appropriate grind size).
8. Load the portafilter with the ground coffee and use the espresso coffee stirrer to stir the ground coffee, and use the tamper to tamp the ground coffee.
9. Confirm that the chamber is properly assembled and that the airflow path is not obstructed.

Top Down Test Steps

Pre-Test Checks

1.0 Espresso Machine / ECU power check

Objective: Verify the espresso machine and ECU power on correctly.
Procedure: Turn on system power.
Expected result: All devices power on without errors. Indicator LEDs, and display are working.
Pass/Fail: __________
Comments: __________

1.1 Pre-Vent Check

Objective: Verify the 3 fans operate correctly before testing.
Procedure:

- With a press of a button all 3 fans turn on for 30 seconds.
- Observe airflow direction and operation

Expected Result:

- Intake and exhaust fans move air through the chamber as intended.
- Internal fan circulates air inside the chamber
- No unusual vibration, noise, or fan stoppage occurs.

Pass/Fail: __________
Comments: __________

1.2 Espresso Machine Temperature Check

Objective: Confirm the machine is at brew-ready temperature (93 Degrees Celsius)
Procedure: Check machine temperature from the Gaggiuino display.
Expected Result: Temperature is within acceptable brew range defined by the team.
Measured Value: __________
Pass/Fail: __________
Comments: __________

1.3 Hardware Pre-Check

Objective: Confirm all hardware is connected.
Procedure: Inspect wiring, mounts, chamber seals, fan mounts, sensors, and scale placement.
Expected Result: No loose wiring, missing parts, disconnected hardware, or damage.
Pass/Fail: __________
Comments:__________

1.4 Brew Enable Check

Objective: Verify the system allows brewing.
Procedure: Start a brew cycle.
Expected Result: The brew cycle begins successfully and the system operates normally.
Pass/Fail: __________
Comments: __________

1.5 Data Logging Check

Objective: Verify system data is recorded during the shot.
Procedure: Pull a shot and observe whether sensor readings, timestamps, and other test data are logged.
Expected Result: Data is continuously recorded during the brew without interruption.
Pass/Fail: __________
Comments: __________

1.6 Network Communications Check

Objective: Verify wireless communication between ESP32 devices and the computer display.
Procedure:

- Start system comms
- Confirm data from the ECU ESP32 is transmitted to the second ESP32 and PC.
- Observe the live data display

Expected Result:

- Wireless communication established
- Data updates appear correctly on the computer screen.
- No major dropouts or comms failures occur during the test

Pass/Fail: __________
Comments: __________

1.7 Circulation Fan Check

Objective: Verify airflow control during the active shot.
Procedure: Observe fan operation while the shot is running.
Expected Result: Fans remain on and maintain expected airflow during the test.
Pass/Fail:__________
Comments: __________

1.8 Scale Enable Check

Objective: Verify the scale connects to Gaggiuino via Bluetooth and provides valid shot weight data.
Procedure: Observe the scale before and during the shot.
Expected Result: The scale responds correctly, tracks weight change, and outputs readable data.
Pass/Fail: __________
Comments: __________

1.9 Visual Test

Objective: Confirm correct overall system operation during the shot.
Procedure: Observe the machine, chamber, fan behavior, sensor, and live display during brewing.
Expected Result:

- Brew occurs normally
- No leaks, disconnections, or unsafe behavior are observed (Although small amounts of splashing may be normal).
- The chamber remains intact
- Displayed data appears reasonable

Pass/Fail:__________

Comments: __________

Post-Test Checks

2.0 Fan Post-Vent Check

Objective: Verify post-test ventilation clears the chamber after the shot.
Procedure: Leave the intake and exhaust fans running for the required post-shot vent period (30 seconds).
Expected Results: The chamber continues ventilating after shot completion for the planned duration.
Pass/Fail: __________
Comments: __________

2.1 Data Validity Check

Objective: Confirm recorded data is reasonable and usable.
Procedure: Review collected logs after the test.
Expected Result: The data file is complete, timestamps are in order, and values are within expected ranges.
Pass/Fail:__________
Comments: __________

2.2 Shot Weight Confirm

Objective: Confirm measured shot weight is 36 grams.
Procedure: Compare the scale reading to the expected shot weight of 36 grams.
Expected Result: Final shot weight is within 5% tolerance.
Measured Value: __________
Pass/Fail: __________
Comments: __________

2.3 Shot Quality Check

Objective: Confirm the espresso shot is acceptable.
Procedure: Inspect crema, flow, shot appearance, and taste.
Expected Result: The shot completes with visual quality and taste quality.
Pass/Fail: __________
Comments: __________

2.4 Hardware Post-Check

Objective: Confirm hardware remains functional after the test.
Procedure: Inspect the system after the brew and vent cycle.
Expected Result: No overheated parts, disconnected wires, leaks, broken mounts, or other hardware issues.
Pass/Fail: __________
Comments: __________

Post-Test Teardown

1. Stop brew operation
2. Shut down fans after the post-vent period is complete.
3. Save all recorded data files.
4. Power off electronics and the espresso machine.
5. Disconnect test equipment.

6. Clean the chamber, scale, empty the porta filter and wash in the sink or with a paper towel, and wipe the sensors with a paper towel if splash was present.
7. Record any failures, unexpected behavior, or follow-up actions.

Conclusions / Discussion

Overall Results: ___________

- ☐ Pass
- ☐ Partial Pass
- ☐ Fail

Summary Notes:

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Requirements Verified:

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Requirements Not Met / Issues Found:

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